

Re-processing Perturb-Seq datasets from FASTQ

Unified Pipeline and Infrastructure Challenges

Kirill Tsukanov
Senior Full Stack Developer & Data Engineer
`ktsukanov@ebi.ac.uk`

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Why re-process from FASTQ?

- ▶ **Consistency:** Most scPerturb datasets are provided as processed H5AD files, but they use different pipelines (Cell Ranger, STARsolo, kallisto) and different reference versions.
- ▶ **Modality Alignment:** Ensuring that the layout and metadata of h5ad files are consistent across studies. This reduces the need for per-dataset adjustments during manual curation.
- ▶ **Quality Control:** Direct access to raw reads would allow for uniform filtering and normalization across the entire Perturbation Catalogue.
- ▶ **Expectations:** When we talked to people, most of them *expect* that data will be fully harmonised in the catalogue.

Data Sourcing: The ENA vs. SRA Dilemma

- ▶ **The Issue:** For direct ENA downloads of single-cell data, it only hosts the cDNA reads for historical reasons.
- ▶ **The Consequence:** Technical reads (Cell Barcodes and UMIs) are missing from ENA FASTQ files.
- ▶ **The Solution:** We must fetch data from the **SRA (Sequence Read Archive)** using the SRA Toolkit.
- ▶ **Workflow:**
 1. Query ENA API for run accessions (SRR).
 2. Use `prefetch` to download SRA files.
 3. Use `fasterq-dump` with `--include-technical`.

Infrastructure and Storage Constraints

- ▶ **Limited Quota:** Only 20 TB allocated on the cluster storage.
- ▶ **Space-Heavy Process:**
 - ▶ fasterq-dump produces **uncompressed** FASTQ files.
 - ▶ Uncompressed original FASTQ files can easily exceed the quota during the download phase, even before we get to the actual analysis.
- ▶ **Storage Conscious Approach:**
 - ▶ Immediate gzip after every given fasterq-dump process completes.
 - ▶ Nextflow cleanup enabled to purge intermediate results.
 - ▶ Threaded downloads and processing to minimize "idle" storage usage.

Unified Pipeline Architecture

- ▶ **Framework:** Nextflow: scales well, works out of the box on the EBI Slurm cluster.
- ▶ **Engine:** kb-python (Bustools + Kallisto wrapper) for mapping.
- ▶ **Environment:** Singularity container, as it is the only solution that works properly on our SLURM cluster.

Pipeline Flow:

1. **Index Building:** Simultaneous cDNA (Ensembl) and KITE (CRISPR feature) indices.
2. **Counting:** Parallel kb count for both modalities across all SRR runs.
3. **Merging:** Per-SRR alignment of cells (matching barcodes).
4. **Concatenation:** Efficient on-disk concatenation (`anndata.experimental`) of all runs into a single `experiment_final.h5ad`.

Handling Dual Modalities (cDNA + CRISPR)

- ▶ The pipeline runs two parallel workflows per SRR:
 1. **Standard Workflow:** Maps cDNA reads to the transcriptome.
 2. **KITE Workflow:** Maps Feature Barcode reads to a guide whitelist (`features.tsv`).
- ▶ **Merging Logic:**
 - ▶ We use the cDNA cell barcodes as the "truth" set.
 - ▶ Guide counts are aligned to these barcodes.
 - ▶ Final H5AD stores gene counts in `.X` and guide counts in `.obsm['guides']`.
- ▶ **Auto-detection:** The pipeline automatically identifies R1 (barcode+UMI) and R2 (read) based on sequence length, simplifying run configuration.

Pipeline Execution Example

```
1 nextflow run main.nf \  
2   -profile slurm,singularity \  
3   --fastq_dir $HPS_PATH/perturb_seq_fastq/SAMN40972597 \  
4   --outdir results \  
5   --chemistry 10xv3 \  
6   --transcriptome_fa $REF/Homo_sapiens.GRCh38.dna.  
   primary_assembly.fa.gz \  
7   --gtf $REF/Homo_sapiens.GRCh38.111.gtf.gz \  
8   --features_tsv ./features.tsv
```

- ▶ Automated SLURM resource allocation.
- ▶ Built-in retry mechanism for transient cluster errors.
- ▶ h5repack with GZIP compression used for final archival.

Case Study: Nadig 2025 (Jurkat)

- ▶ **Data Volume:**
 - ▶ Total Compressed FASTQ: ~ 2 TB.
 - ▶ Scale: 2,688 FASTQ files across multiple SRR runs.
- ▶ **Performance (Nadig 2025 Jurkat):**
 - ▶ **Download:** ~ 3 hours.
 - ▶ **Processing:** ~ 5 hours.
 - ▶ **Archival:** ~ 1 hour (compression + upload).
- ▶ **Future Scaling:**
 - ▶ These times will be further improved.
 - ▶ We aim for a very fast, resilient pipeline that scales for the entire catalogue.

Next Steps

- ▶ **Validation:** Perform correlation analysis between our re-processed counts and the authors' original H5AD files for a basic sanity check.
- ▶ **Scaling:** Extend processing to other large datasets.
- ▶ **Refinement:** Automate the generation of `features.tsv` from supplementary tables for more studies.
- ▶ **Integration:** Feed the unified H5ADs into our downstream pipelines: curation, differential expression, GSEA.

Questions & Discussion

- ▶ **Post-processing:** Any recommended additional filtering or normalization steps for unified H5AD files?
- ▶ **Caveats:** What are the most common pitfalls or "gotchas" when re-processing Perturb-Seq from FASTQ?
- ▶ **Improvements:** Any other suggestions for enhancing the pipeline or downstream interpretation?

Thank you for your feedback!